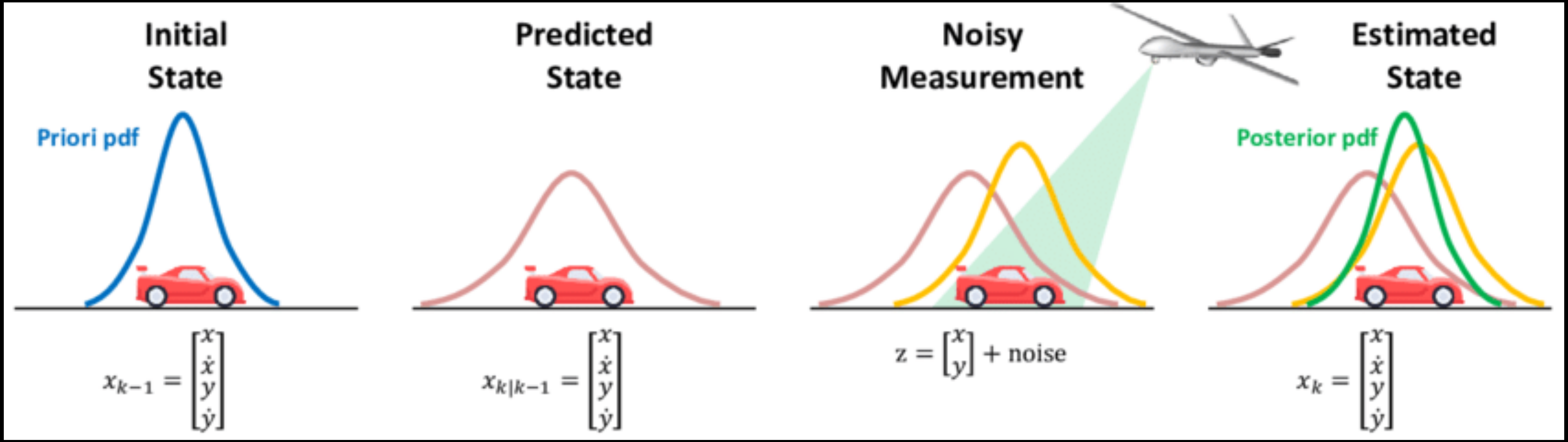
Kalman Filter

State estimation



Problem set up

Want to make sure $|x_t - \hat{x}_t|$ is small as possible.

Set up for Gaussian Linear Dynamics

- Gaussian linear dynamics

White noise

$$x_{t+1} = Ax_t + Bu_t + w_t, \qquad \mathbb{E}[w_t w_t^T] = Q, \qquad w_t \sim \mathcal{N}(0, Q)$$

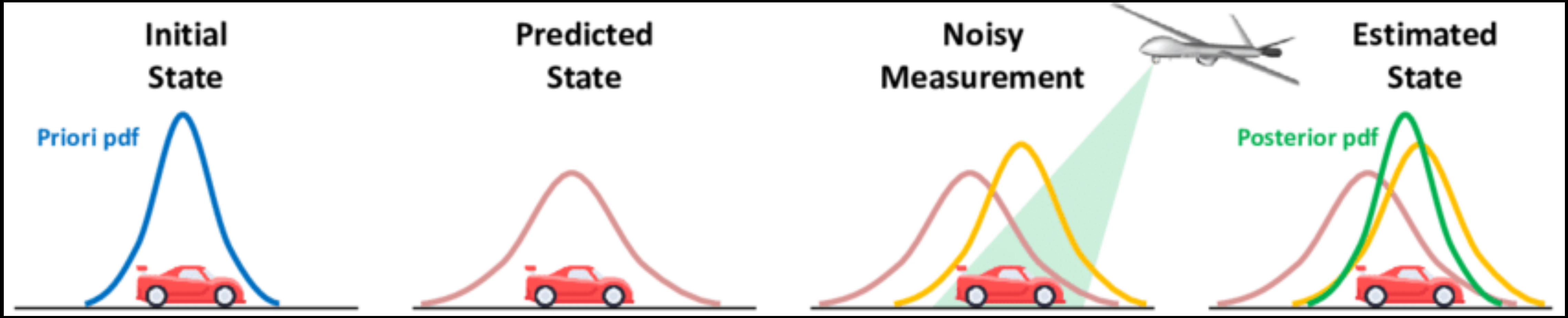
$$y_t = Cx_t + v_t, \qquad \mathbb{E}[v_t v_t^T] = R, \qquad v_t \sim \mathcal{N}(0, R)$$

- Estimate of current state

$$\hat{x}_t \sim p(\hat{x}_t \mid u_{0:t-1}, y_{1:t}, x_0) \qquad \hat{x}_t \sim \mathcal{N}(\mu_t, P_t)$$

- Covariance of error $P_t = \mathbb{E}[e_t e_t^T] = \mathbb{E}[(x_t - \hat{x}_t)(x_t - \hat{x}_t)^T]$
- What is μ_t, P_t ? Depend on previous values.

Outline of steps



1. Get previous estimate
2. **Predict** new estimate (prior)
3. Receive measurement
4. Update estimate given measurement (posterior)

Predict step

- Given μ_{t-1}, P_{t-1} , estimate at previous time step
- Predict next state estimate using **noise-free** dynamics only

$$\mu_t^p = A\mu_{t-1} + Bu_{t-1} \qquad P_t^p = AP_{t-1}A^T + Q$$

Receive measurement

- Receive measurement & compare with predicted measurement

$$h_t = y_t - C\hat{x}_t^p$$

Update prediction

- Update mean give prediction and measurement
$$\mu_t = \mu_t^p + K_t(y_t - C\mu_t^p)$$
- Update error covariance $P_t = (I - K_tC)P_t^p(I - K_tC)^T + K_tRK_t^T$

Choosing the best gain

- How to choose K_t ?
  - The one that minimizes $\mathrm{Tr}(P_t)$ (mean square error)

… some algebra… $K_t = P_t^p C^T (C P_t^p C^T + R)^{-1}$

- Final error covariance update

$$P_t = (I - K_t C) P_t^p$$

Kalman Filter algorithm

Start with initial state estimate $\hat{x}_0 \sim \mathcal{N}(\mu_0, P_0)$

For t = 1, 2, …

1. Predict step $\mu_{t-1}, P_{t-1} \rightarrow \mu_t^p, P_t^p$

   $\mu_t^p = A\mu_{t-1} + Bu_{t-1}$ $\qquad P_t^p = AP_{t-1}A^T + Q$

2. Compute Kalman gain $K_t = P_t^p C^T (C P_t^p C^T + R)^{-1}$
3. Get measurement
4. Update step $\mu_t^p, P_t^p \rightarrow \mu_t, P_t$

   $\mu_t = \mu_t^p + K_t(y_t - C\mu_t^p)$ $\qquad P_t = (I - K_t C)P_t^p$

- Iterate *forward* in time

Do these equations look familiar?

$$P_t^p = AP_{t-1}A^T + Q \qquad K_t = P_t^p C^T (C P_t^p C^T + R)^{-1}$$

From LQR

$$P_{t-1} = Q + A^T P_t A - A^T P_t B (B^T P_t B + R)^{-1} B^T P_t A$$
$$K_t = (B^T P_{t+1} B + R)^{-1} B^T P_{t+1} A$$

Do these equations look familiar?

$$P_t^p = AP_{t-1}A^T + Q \qquad K_t = P_t^p C^T (CP_t^p C^T + R)^{-1}$$

- Kind of…but not quite
- Some more algebra.
- Sub $P_{t-1} = (I - K_{t-1}C)P_{t-1}^p$ into $P_t^p = AP_{t-1}A^T + Q$
- We get…

$$P_{t+1}^p = Q + AP_t^p A^T - AP_t^p C^T (CP_t^p C^T + R)^{-1} CP_t^p A^T$$

From LQR

$$P_{t-1} = Q + A^T P_t A - A^T P_t B(B^T P_t B + R)^{-1} B^T P_t A$$

$$K_t = (B^T P_{t+1} B + R)^{-1} B^T P_{t+1} A$$

Duality!

LQR Description	LQR	Kalman filter	KF Description
Dynamics	A	A^T	Dynamics transposed
Control matrix	B	C^T	Observation matrix transposed
State cost matrix	Q	Q	Process noise covariance
Control cost matrix	R	R	Measurement noise covariance
Terminal cost matrix	Q_N	P_0	Initial state covariance
Value function	P_t	P_t^p	Predicted error covariance